

Supplementary Material for “Conformal Prediction Regions for Compound-Gaussian Radar Clutter: Exact Laws, Thermal Noise, and the Limits of Onset Detection”

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All numbers below are generated from the saved outcomes in the project repository (`research/r7c`). Six protocols were frozen and hash-recorded before their outcomes were computed: the main IPIX study (`PROTOCOL.md`), single-pulse detection (`detection/PROTOCOL_DETECTION.md`), onset (`detection/PROTOCOL_ONSET.md`), baselines (`baselines/PROTOCOL_BASELINES.md`), the second radar (`jku/PROTOCOL_JKU.md`) and classical detectors, gradual emergence and ACI contamination (`r10/PROTOCOL_R10.md`). A seventh (`sdrdsp/PROTOCOL_SDRDSP.md`) was frozen before the SDRDSP data were obtained.

1 Data exposure

Table 1 states which data each stage used, so that the strength of each validation claim can be judged. Only the hold-out series and the second radar were not used while the methods were being developed.

Table 1: Data exposure by stage. “Seen” means that outcomes on these data were computed while methods, orders or comparisons could still change.

Data	Used for	Status of the reported results
IPIX, 14 sessions, rotation 1	Method development (R1–R7): method list, model orders, comparisons	Seen; not reported
IPIX, same sessions, rotations 2–3	Main study, detection, onset, baselines, classical detectors (R7–R10)	Frozen-protocol evaluation, computed once per protocol; the recordings were seen in rotation 1, so this is not an untouched confirmation
IPIX, day transfer	Transfer and ACI analyses	Frozen-protocol evaluation on the same recordings
IPIX #269, #287	Hold-out	Hashed and unopened until the main analysis was complete; evaluated once (small test sets)
JKU 77 GHz, reference run	Second-radar study	Not used in development; frozen protocol; the general-form gain curve is post hoc
SDRDSP (pending)	Untouched sea-clutter confirmation	Protocol frozen and hashed before the data were obtained

2 Hold-out series

Two further McMaster series (#269 and #287, the widely used “high” and “low” sea-state files, one provider-preprocessed VV cell each, from the same 1993 campaign) were hashed and left unopened until the main analysis was complete, and then evaluated once (Table 2). They confirm the direction of the main result, with larger gains. They also show a case in which the classical Gaussian threshold over-covers while conformal IN-ARCP stays near nominal. Their test sets are small (about 165 episodes per unit), so the hold-out is a directional check. Mondrian calibration returns infinite regions at $1 - \alpha = 0.99$ there, because each bin receives fewer calibration episodes than the rank requires.

Table 2: Hold-out series #269 and #287 ($1 - \alpha = 0.90$; six units per column).

Method	$m = 8$		$m = 16$	
	Radius ratio	Coverage (min)	Radius ratio	Coverage (min)
Noise-aware AR(4) (proposed)	0.454	0.890 (0.816)	0.498	0.893 (0.847)
Noise-aware AR(2) (proposed)	0.541	0.887 (0.828)	0.533	0.895 (0.859)
AR(4), innovation-normalized	0.546	0.908 (0.847)	0.515	0.906 (0.883)
AR(2), innovation-normalized	0.582	0.905 (0.840)	0.572	0.906 (0.871)
Noise-aware AR(1)	0.921	0.898 (0.840)	0.921	0.903 (0.859)
IN-ARCP, AR(1)	1.000	0.906 (0.853)	1.000	0.910 (0.859)
Gaussian plug-in of IN-ARCP	1.178	0.949 (0.896)	1.104	0.945 (0.895)
Mondrian IN-ARCP	1.000	0.895 (0.804)	0.981	0.896 (0.810)
Learned-scale CP	0.629	0.893 (0.804)	0.662	0.919 (0.791)
Invariant MLP CP	1.809	0.884 (0.736)	2.158	0.903 (0.761)
Raw-RMS normalized CP	1.321	0.880 (0.706)	1.243	0.887 (0.706)
Unnormalized CP	1.142	0.868 (0.607)	1.161	0.859 (0.589)
Gaussian plug-in of noise-aware AR(4)	0.411	0.866 (0.816)	0.472	0.856 (0.755)

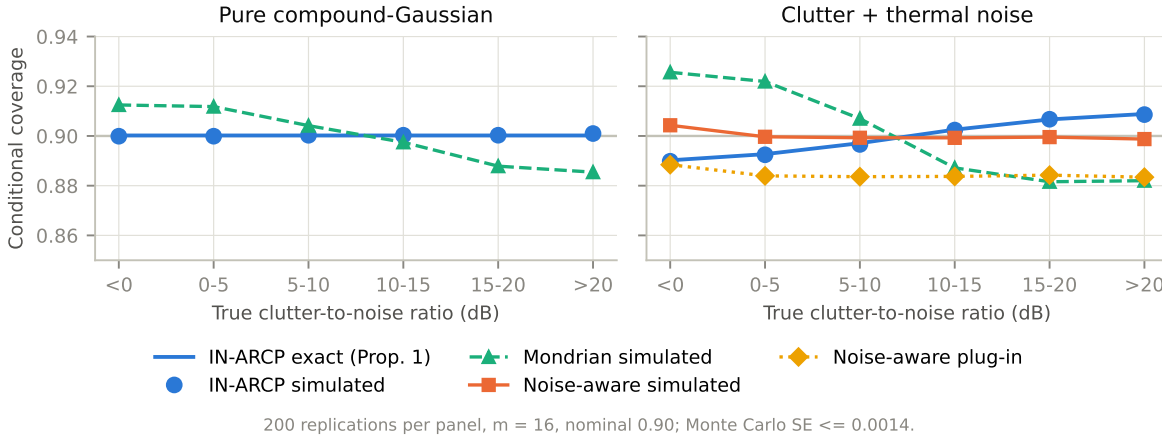


Figure 1: Synthetic verification: coverage conditional on the true clutter-to-noise ratio. The line is the exact law of Proposition 1; markers are simulation.

3 Synthetic verification

4 Proofs of Propositions 2 and 5

Equation and proposition numbers refer to the paper.

Proposition 2. If $e \sim \mathcal{CN}(0, v)$ and a Swerling-1 target of power SP is added, a threshold test on $|e|^2/v$ at false-alarm probability α detects with probability $\alpha^{1/(1+SP/v)}$, while the power-only test on $|Y|^2/P$ detects with $\alpha^{1/(1+S)}$; equal P_d therefore requires SCRs in the ratio $G = P/v$. For the centre rH_m under (1) with AR(1) speckle, $v = \nu\{c(1 + |r|^2 - 2\Re\bar{r}\rho) + 1 + |r|^2\}$, which gives (5). For the optimal predictor from the infinite past, v is the one-step prediction-error variance of the AR(1)-plus-white-noise process, $\exp\{\int \log f\}$ by the Kolmogorov–Szegő formula (Brockwell and Davis, *Time Series: Theory and Methods*, 1991). With $f(\omega) = \nu\{c(1 - |\rho|^2) + |1 - \rho e^{-j\omega}|^2\}/|1 - \rho e^{-j\omega}|^2$, the denominator contributes zero, and the numerator factorizes as $\nu K|1 - \beta' e^{-j\omega}|^2$ with $K(1 + |\beta'|^2) = \beta$, $K\beta' = \rho$ and $|\beta'| < 1$, which gives $v = \nu K$ and (6). Because the optimal predictor's error variance is at most that of any linear predictor, including 0 and rH_m , $G_{\text{opt}} \geq \max\{1, G_{\text{IN}}\}$.

Proposition 5. Write $t = T(Z_H)$, so that the observed statistic is $T(X_H) = \sigma t$ and $S(X) = S(Z)$. For bin b let $w_b(t) = \mathbb{P}\{\sigma t \in B_b\}$. Since σ is independent of Z , for every x

$$\mathbb{P}\{S \leq x, \sigma t \in B_b\} = \mathbb{E}[\varphi_x(t) w_b(t)], \quad (\text{S1})$$

where $\varphi_x(t) = \mathbb{P}\{S \leq x | t\}$. (i) If S is independent of t , then $\varphi_x \equiv F_S(x)$, every bin-conditional law of S equals F_S , all q_b equal $F_S^{-1}(1 - \alpha)$ and $\mathcal{C}(\sigma) = 1 - \alpha$.

(ii) Let $x_0 = F_S^{-1}(1 - \alpha)$. For the top bin, $w_K(t) = \mathbb{P}\{\sigma \geq e_{K-1}/t\}$ is nondecreasing in t . Because T is continuous, positive and homogeneous and Z_H has full support, t has full support on $(0, \infty)$; as e_{K-1}/t then ranges over $(0, \infty)$ and G

is non-degenerate, $w_K(t)$ is not almost surely constant. By assumption φ_{x_0} is strictly increasing. Let t' be an independent copy of t , $\Delta_\varphi = \varphi_{x_0}(t) - \varphi_{x_0}(t')$ and $\Delta_w = w_K(t) - w_K(t')$. Then

$$\text{Cov}\{\varphi_{x_0}(t), w_K(t)\} = \frac{1}{2} \mathbb{E}[\Delta_\varphi \Delta_w] > 0,$$

because $\Delta_\varphi \Delta_w \geq 0$, with strict inequality on the event $\{w_K(t) \neq w_K(t')\}$, which has positive probability. By (S1), $\mathbb{P}\{S \leq x_0 \mid \sigma t \in B_K\} = \mathbb{E}[\varphi_{x_0} w_K] / \mathbb{E}[w_K] > \mathbb{E}[\varphi_{x_0}] = 1 - \alpha$. The bin-conditional distribution function of S is continuous, so there is $x_1 < x_0$ at which it still exceeds $1 - \alpha$; hence $q_K \leq x_1 < x_0$ and, F_S being strictly increasing, $F_S(q_K) < 1 - \alpha$. For the bottom bin, $w_1(t) = \mathbb{P}\{\sigma < e_1/t\}$ is nonincreasing, the covariance is negative, and the same steps give $q_1 > x_0$ and $F_S(q_1) > 1 - \alpha$.

The coverage at texture σ is $\mathcal{C}(\sigma) = \sum_b \mathbb{P}\{S \leq q_b, \sigma t \in B_b\}$. As $\sigma \rightarrow \infty$, $\sigma t \rightarrow \infty$ almost surely, so $\mathbf{1}\{\sigma t \in B_K\} \rightarrow 1$ and, by dominated convergence, $\mathcal{C}(\sigma) \rightarrow F_S(q_K)$; as $\sigma \rightarrow 0$, $\mathcal{C}(\sigma) \rightarrow F_S(q_1)$. An unbinned threshold q gives $\mathcal{C}(\sigma) = F_S(q)$ for every σ .

Finally, for IN-ARCP at $r = \rho$ and $T = s_\rho$, $S = |E|/t$ with the innovation E independent of Z_H , so $\varphi_x(t) = \mathbb{P}\{|E| \leq xt\}$ is strictly increasing in t , which is condition (ii). $\blacksquare$

5 Proof sketch of Proposition 4 (zero-noise limit)

With $\gamma = \nu c$ the history log-likelihood converges locally uniformly on $\gamma > 0$ to $-m \log \gamma - \gamma^{-1} h^H R_H^{-1} h$, whose unique maximizer is $h^H R_H^{-1} h / m$; coercivity at 0 and ∞ gives convergence of the maximizers. The Markov property gives $\mu \rightarrow \rho h_m$, and $(1 - |\rho|^2) R_H^{-1} = W_\rho^H W_\rho$, with W_ρ the bidiagonal whitening matrix, identifies the limiting predictive variance with $s_\rho(h)^2$.

6 Day transfer and texture-proxy sensitivity

Table 3: Day transfer ($m = 16$, $1 - \alpha = 0.90$): fit and calibrate on other days, test on each of six held-out days.

Method	Mean coverage	Worst held-out day	Radius ratio
Noise-aware AR(4) (proposed)	0.898	0.863	0.936
Noise-aware AR(2) (proposed)	0.898	0.883	0.956
AR(4), innovation-normalized	0.901	0.894	0.993
AR(2), innovation-normalized	0.899	0.891	1.024
Noise-aware AR(1)	0.900	0.882	1.037
IN-ARCP, AR(1)	0.903	0.882	1.000
Gaussian plug-in of IN-ARCP	0.906	0.893	1.006
Mondrian IN-ARCP	0.907	0.881	0.984
Learned-scale CP	0.898	0.774	0.967
Invariant MLP CP	0.880	0.622	1.200
Raw-RMS normalized CP	0.906	0.681	1.466
Unnormalized CP	0.909	0.716	1.470
Gaussian plug-in of noise-aware AR(4)	0.876	0.842	0.893

Texture-proxy sensitivity (pre-specified windows; mean maximum quintile deviation at $1 - \alpha = 0.90$, $m = 16$): with ± 512 pulses IN-ARCP 0.035, NA-AR(4) 0.043, unnormalized 0.181, neural centre 0.103; with ± 2048 pulses IN-ARCP 0.033, unnormalized 0.141. The ranking is the same as with ± 1024 .

7 Additional figures for the IPIX study

8 Detection study: all levels and methods

Pooled detection probability by SCR, measured false-alarm rates, exact-law predictions and SCR gains with 95% day-cluster intervals (output of `detection/analyze_detection.py`):

units: 56; days: 6; test episodes: 74980

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=== 1-alpha = 0.9 (Swirling 1; Pd pooled mean over units; SCR in dB rel. local clutter+noise power) ===
method Pfa_meas -5.0 -2.5 0.0 2.5 5.0 7.5 10.0 12.5 15.0 17.5 20.0 22.5 25.0
CAlloc 0.1030 0.166 0.219 0.307 0.433 0.579 0.711 0.814 0.888 0.934 0.963 0.979 0.988 0.993
CA16 0.1002 0.242 0.312 0.406 0.516 0.631 0.736 0.823 0.887 0.931 0.959 0.977 0.987 0.992
U 0.1011 0.446 0.549 0.651 0.745 0.823 0.883 0.926 0.955 0.974 0.985 0.992 0.995 0.997
RMS 0.1007 0.442 0.541 0.642 0.734 0.813 0.877 0.922 0.953 0.972 0.984 0.991 0.995 0.997
MLP 0.0990 0.447 0.548 0.650 0.743 0.823 0.884 0.927 0.956 0.974 0.985 0.992 0.995 0.997

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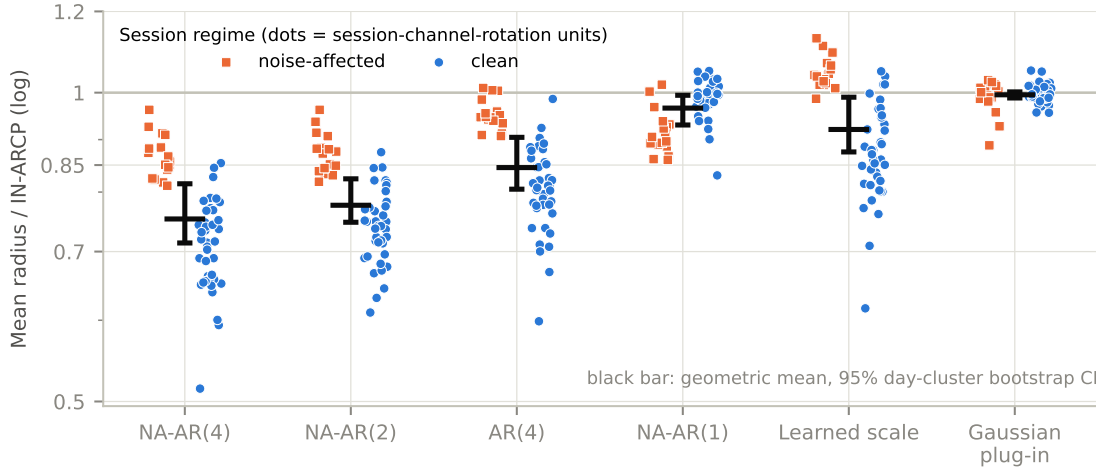


Figure 2: Per-unit radius relative to IN-ARCP at $1 - \alpha = 0.90$, $m = 16$, by session regime (noise-affected: fitted noise power above 1% of mean power).

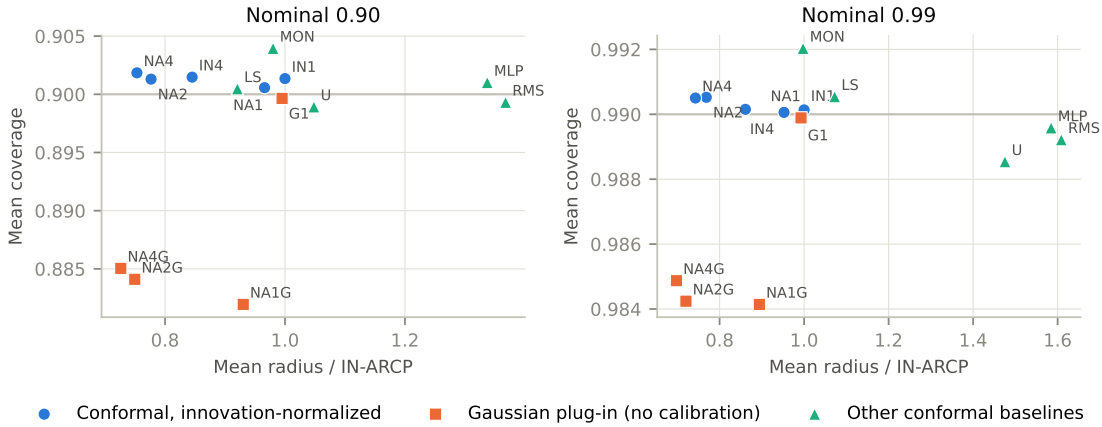


Figure 3: Mean coverage against mean radius (confirmatory, $m = 16$). Gaussian plug-ins of the richer noise-aware models are shorter but under-cover; conformal calibration restores nominal coverage. Labels: NAp and INp , noise-aware and innovation-normalized $AR(p)$ ($IN1$ is IN-ARCP); $G1$ and $NApG$, Gaussian plug-ins; MON , Mondrian; LS , learned scale; U , unnormalized; RMS , raw-RMS; MLP , neural centre.

LS	0.0995	0.534	0.624	0.710	0.784	0.847	0.898	0.935	0.960	0.977	0.987	0.993	0.996	0.997
MON	0.0961	0.540	0.634	0.720	0.794	0.858	0.907	0.941	0.965	0.979	0.988	0.993	0.996	0.998
G1	0.1004	0.542	0.635	0.722	0.795	0.859	0.908	0.942	0.965	0.980	0.989	0.993	0.996	0.998
IN1	0.0987	0.540	0.633	0.720	0.793	0.857	0.906	0.941	0.964	0.979	0.988	0.993	0.996	0.998
IN2	0.0991	0.553	0.643	0.728	0.801	0.862	0.909	0.943	0.966	0.980	0.989	0.993	0.996	0.998
IN4	0.0985	0.574	0.661	0.741	0.811	0.869	0.914	0.946	0.967	0.981	0.989	0.994	0.996	0.998
NA1	0.0994	0.553	0.648	0.735	0.810	0.872	0.917	0.948	0.969	0.982	0.990	0.994	0.997	0.998
NA2	0.0987	0.609	0.692	0.767	0.834	0.887	0.926	0.954	0.972	0.984	0.991	0.995	0.997	0.998
NA4	0.0982	0.614	0.696	0.770	0.836	0.888	0.927	0.954	0.973	0.984	0.991	0.995	0.997	0.998
NA4G	0.1150	0.630	0.710	0.782	0.845	0.895	0.932	0.958	0.975	0.986	0.992	0.995	0.997	0.998
IN1 exact-law prediction 0.507 0.614 0.710 0.791 0.856 0.906 0.942 0.965 0.979 0.988 0.993 0.996 0.998														
exact-law MAE, IN1 Pd: per unit x SCR 0.0140; pooled curve 0.0053														
-- SCR gain (dB) needed for Pd=0.5; positive = less SCR than reference; 95% day-cluster CI --														
IN1	SCR	-3.0 dB		vs CA16	5.06 [3.25, 6.46]		vs Caloc	6.64 [4.50, 8.41]						
IN4	SCR	-3.1 dB		vs CA16	5.19 [3.41, 6.53]		vs Caloc	6.76 [4.69, 8.46]						
NA4	SCR	-3.4 dB		vs CA16	5.41 [3.85, 6.61]		vs Caloc	6.99 [5.10, 8.52]						
G1	SCR	-3.1 dB		vs CA16	5.10 [3.35, 6.44]		vs Caloc	6.67 [4.65, 8.39]						
U	SCR	-2.6 dB		vs CA16	4.61 [2.79, 6.04]		vs Caloc	6.19 [4.15, 8.03]						
MLP	SCR	-2.6 dB		vs CA16	4.66 [2.72, 6.11]		vs Caloc	6.23 [3.87, 8.10]						
LS	SCR	-2.8 dB		vs CA16	4.83 [2.83, 6.34]		vs Caloc	6.40 [4.06, 8.30]						
-- SCR gain (dB) needed for Pd=0.9; positive = less SCR than reference; 95% day-cluster CI --														
IN1	SCR	5.2 dB		vs CA16	7.98 [5.03, 10.43]		vs Caloc	7.93 [5.02, 10.37]						
IN4	SCR	4.3 dB		vs CA16	8.86 [5.48, 11.49]		vs Caloc	8.80 [5.45, 11.44]						
NA4	SCR	3.1 dB		vs CA16	10.05 [6.52, 12.77]		vs Caloc	9.99 [6.51, 12.72]						
G1	SCR	5.1 dB		vs CA16	8.02 [5.09, 10.42]		vs Caloc	7.97 [5.11, 10.38]						
U	SCR	7.1 dB		vs CA16	6.09 [3.33, 8.23]		vs Caloc	6.04 [3.30, 8.21]						
MLP	SCR	7.0 dB		vs CA16	6.12 [3.70, 7.88]		vs Caloc	6.07 [3.70, 7.81]						
LS	SCR	5.4 dB		vs CA16	7.78 [4.32, 10.40]		vs Caloc	7.72 [4.35, 10.37]						
=== 1-alpha = 0.99 (Swelling 1; Pd pooled mean over units; SCR in dB rel. local clutter+noise power) ===														
method	Pfa_meas	-5.0	-2.5	0.0	2.5	5.0	7.5	10.0	12.5	15.0	17.5	20.0	22.5	25.0
Caloc	0.0110	0.022	0.034	0.062	0.125	0.237	0.394	0.566	0.714	0.822	0.893	0.939	0.966	0.980
CA16	0.0103	0.081	0.129	0.203	0.306	0.434	0.568	0.695	0.796	0.872	0.922	0.954	0.974	0.985
U	0.0115	0.131	0.208	0.308	0.428	0.558	0.680	0.782	0.860	0.913	0.948	0.970	0.983	0.990
RMS	0.0108	0.176	0.271	0.383	0.503	0.625	0.733	0.822	0.887	0.931	0.960	0.977	0.987	0.993
MLP	0.0104	0.176	0.272	0.388	0.510	0.632	0.742	0.831	0.893	0.935	0.962	0.978	0.988	0.993

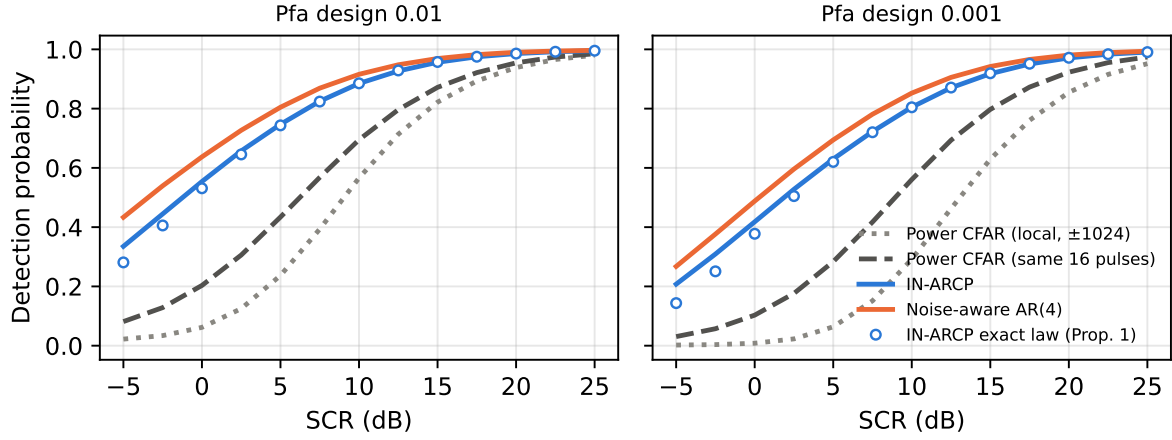


Figure 4: Detection of injected Swerling-1 targets on IPIX at design false-alarm probabilities 0.01 and 0.001.

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LS      0.0095  0.288  0.391  0.495  0.597  0.694  0.776  0.846  0.900  0.938  0.963  0.979  0.988  0.993
MON     0.0080  0.326  0.434  0.546  0.651  0.743  0.822  0.883  0.927  0.956  0.974  0.985  0.992  0.995
G1      0.0101  0.337  0.448  0.558  0.661  0.751  0.828  0.887  0.929  0.957  0.975  0.986  0.992  0.995
IN1     0.0099  0.335  0.445  0.554  0.657  0.747  0.825  0.885  0.927  0.956  0.975  0.986  0.992  0.995
IN2     0.0102  0.357  0.464  0.569  0.670  0.758  0.832  0.889  0.931  0.959  0.976  0.986  0.992  0.996
IN4     0.0098  0.377  0.482  0.586  0.681  0.765  0.836  0.892  0.932  0.960  0.977  0.987  0.992  0.996
NA1     0.0099  0.354  0.466  0.577  0.681  0.773  0.848  0.903  0.940  0.965  0.980  0.989  0.994  0.996
NA2     0.0095  0.423  0.531  0.632  0.723  0.802  0.868  0.915  0.948  0.969  0.982  0.990  0.994  0.997
NA4     0.0095  0.433  0.539  0.637  0.727  0.805  0.869  0.916  0.948  0.969  0.982  0.990  0.994  0.997
NA4G    0.0151  0.465  0.566  0.659  0.745  0.819  0.878  0.921  0.952  0.971  0.984  0.991  0.995  0.997

IN1 exact-law prediction 0.281  0.406  0.531  0.645  0.743  0.824  0.885  0.929  0.957  0.975  0.986  0.992  0.995
exact-law MAE, IN1 Pd: per unit x SCR 0.0210; pooled curve 0.0105
-- SCR gain (dB) needed for Pd=0.5; positive = less SCR than reference; 95% day-cluster CI --
IN1  SCR -0.9 dB | vs CA16 6.96 [ 4.29, 9.04] | vs Caloc 9.81 [ 6.41,12.61]
IN4  SCR -1.3 dB | vs CA16 7.37 [ 4.54, 9.52] | vs Caloc 10.22 [ 6.67,13.26]
NA4  SCR -1.9 dB | vs CA16 8.03 [ 5.45,10.02] | vs Caloc 10.88 [ 7.56,13.66]
G1   SCR -1.0 dB | vs CA16 7.05 [ 4.40, 9.11] | vs Caloc 9.90 [ 6.48,12.71]
U    SCR 3.8 dB | vs CA16 2.30 [ 0.78, 3.44] | vs Caloc 5.15 [ 3.34, 6.61]
MLP  SCR 2.2 dB | vs CA16 3.88 [ 1.89, 5.34] | vs Caloc 6.73 [ 3.89, 8.70]
LS   SCR 0.4 dB | vs CA16 5.72 [ 2.75, 7.96] | vs Caloc 8.57 [ 4.89,11.49]
-- SCR gain (dB) needed for Pd=0.9; positive = less SCR than reference; 95% day-cluster CI --
IN1  SCR 8.7 dB | vs CA16 7.62 [ 4.55,10.01] | vs Caloc 8.97 [ 5.58,11.67]
IN4  SCR 8.0 dB | vs CA16 8.35 [ 4.95,10.96] | vs Caloc 9.70 [ 5.90,12.64]
NA4  SCR 6.5 dB | vs CA16 9.81 [ 6.27,12.45] | vs Caloc 11.16 [ 7.28,14.14]
G1   SCR 8.7 dB | vs CA16 7.67 [ 4.71,10.02] | vs Caloc 9.02 [ 5.70,11.73]
U    SCR 13.5 dB | vs CA16 2.80 [ 1.01, 4.34] | vs Caloc 4.15 [ 2.23, 5.73]
MLP  SCR 12.0 dB | vs CA16 4.36 [ 2.30, 5.85] | vs Caloc 5.71 [ 3.23, 7.35]
LS   SCR 10.2 dB | vs CA16 6.14 [ 2.75, 8.69] | vs Caloc 7.49 [ 3.78,10.46]

=== 1-alpha = 0.999 (Swerling 1; Pd pooled mean over units; SCR in dB rel. local clutter+noise power) ===
method Pfa_meas -5.0 -2.5 0.0 2.5 5.0 7.5 10.0 12.5 15.0 17.5 20.0 22.5 25.0
Caloc  0.0008  0.002  0.004  0.008  0.023  0.064  0.151  0.292  0.463  0.629  0.761  0.855  0.915  0.952
CA16    0.0007  0.031  0.057  0.103  0.176  0.284  0.417  0.561  0.693  0.797  0.873  0.923  0.955  0.974
U        0.0014  0.023  0.048  0.092  0.159  0.251  0.362  0.485  0.606  0.710  0.796  0.863  0.911  0.944
RMS      0.0007  0.057  0.108  0.184  0.287  0.411  0.539  0.665  0.772  0.853  0.910  0.946  0.969  0.982
MLP      0.0009  0.057  0.110  0.190  0.297  0.426  0.559  0.683  0.788  0.865  0.918  0.952  0.972  0.984
LS       0.0010  0.124  0.198  0.290  0.390  0.494  0.596  0.693  0.778  0.849  0.902  0.940  0.964  0.979
MON      0.0000  0.000  0.000  0.000  0.000  0.000  0.000  0.000  0.000  0.000  0.000  0.000  0.000  0.000
G1       0.0013  0.226  0.333  0.446  0.560  0.665  0.756  0.834  0.893  0.934  0.961  0.977  0.987  0.993
IN1      0.0011  0.208  0.309  0.418  0.528  0.630  0.723  0.805  0.870  0.918  0.951  0.971  0.984  0.991
IN2      0.0009  0.209  0.309  0.418  0.526  0.629  0.722  0.803  0.868  0.916  0.950  0.970  0.983  0.990
IN4      0.0008  0.230  0.332  0.440  0.543  0.641  0.729  0.804  0.866  0.913  0.947  0.969  0.982  0.990
NA1      0.0012  0.219  0.325  0.440  0.554  0.661  0.758  0.837  0.897  0.937  0.963  0.979  0.988  0.993
NA2      0.0009  0.272  0.380  0.491  0.598  0.695  0.783  0.853  0.906  0.943  0.966  0.981  0.989  0.994
NA4      0.0007  0.267  0.377  0.488  0.596  0.694  0.781  0.852  0.905  0.942  0.966  0.981  0.989  0.994
NA4G     0.0028  0.370  0.477  0.581  0.674  0.760  0.833  0.890  0.931  0.958  0.976  0.986  0.992  0.996

IN1 exact-law prediction 0.144  0.251  0.378  0.505  0.620  0.720  0.804  0.871  0.919  0.951  0.972  0.984  0.991
exact-law MAE, IN1 Pd: per unit x SCR 0.0266; pooled curve 0.0155
-- SCR gain (dB) needed for Pd=0.5; positive = less SCR than reference; 95% day-cluster CI --
IN1  SCR 1.9 dB | vs CA16 6.84 [ 4.08, 8.94] | vs Caloc 10.95 [ 7.28,13.95]
IN4  SCR 1.7 dB | vs CA16 7.07 [ 4.04, 9.44] | vs Caloc 11.19 [ 7.42,14.41]
NA4  SCR 0.5 dB | vs CA16 8.27 [ 5.76,10.32] | vs Caloc 12.39 [ 8.88,15.43]
G1   SCR 1.2 dB | vs CA16 7.52 [ 5.19, 9.39] | vs Caloc 11.64 [ 8.28,14.48]
U    SCR 10.4 dB | vs CA16 -1.64 [-4.25,-0.30] | vs Caloc 2.48 [ 0.60, 3.74]
MLP  SCR 6.2 dB | vs CA16 2.55 [ 1.07, 3.82] | vs Caloc 6.67 [ 3.97, 8.33]
LS   SCR 5.1 dB | vs CA16 3.69 [ 1.34, 5.58] | vs Caloc 7.81 [ 4.35,10.22]
-- SCR gain (dB) needed for Pd=0.9; positive = less SCR than reference; 95% day-cluster CI --
IN1  SCR 11.5 dB | vs CA16 7.22 [ 4.16, 9.55] | vs Caloc 9.57 [ 5.97,12.67]
IN4  SCR 11.2 dB | vs CA16 7.48 [ 4.09,10.22] | vs Caloc 9.90 [ 6.03,13.47]
NA4  SCR 9.7 dB | vs CA16 8.96 [ 6.01,11.33] | vs Caloc 11.37 [ 7.84,14.47]
G1   SCR 10.8 dB | vs CA16 7.88 [ 5.23,10.04] | vs Caloc 10.22 [ 6.92,13.23]
U    SCR 18.6 dB | vs CA16 0.08 [-1.16, 0.98] | vs Caloc 2.35 [ 0.65, 3.73]
MLP  SCR 15.9 dB | vs CA16 2.85 [ 1.26, 4.16] | vs Caloc 5.32 [ 3.01, 6.76]
LS   SCR 14.6 dB | vs CA16 4.10 [ 1.22, 6.30] | vs Caloc 6.51 [ 2.89, 9.48]

```

9 Persistent targets after onset

Frozen onset study (detection/PROTOCOL_ONSET.md): per-look detection probability, cumulative detection within K looks, and clean cumulative false-alarm probabilities:

```

units 56; segments 74980

=== per-look Pfa design 0.01 ===
clean cumulative false-alarm prob, K=1,2,4,8: IN1  0.0099 0.0189 0.0357 0.0683

```

```

clean cumulative false-alarm prob, K=1,2,4,8: G1      0.0101 0.0192 0.0369 0.0706
clean cumulative false-alarm prob, K=1,2,4,8: IN4     0.0098 0.0191 0.0369 0.0721
clean cumulative false-alarm prob, K=1,2,4,8: NA4     0.0095 0.0182 0.0363 0.0701
clean cumulative false-alarm prob, K=1,2,4,8: CA16    0.0103 0.0170 0.0302 0.0540
clean cumulative false-alarm prob, K=1,2,4,8: CAlc    0.0113 0.0147 0.0209 0.0308
-- per-look Pd at SCR 10 dB, Doppler random (looks j=0..7) --
IN1  0.885 0.711 0.639 0.478 0.084 0.012 0.005 0.002
G1   0.887 0.715 0.645 0.497 0.068 0.011 0.005 0.003
IN4  0.891 0.725 0.563 0.256 0.022 0.003 0.001 0.001
NA4  0.914 0.883 0.705 0.571 0.431 0.287 0.201 0.178
CA16 0.691 0.601 0.455 0.254 0.100 0.039 0.017 0.009
CAlc 0.562 0.564 0.564 0.563 0.565 0.565 0.563 0.565
-- per-look Pd at SCR 10 dB, Doppler matched (looks j=0..7) --
IN1  0.885 0.024 0.013 0.007 0.003 0.002 0.002 0.001
G1   0.887 0.025 0.014 0.008 0.004 0.003 0.002 0.002
IN4  0.891 0.049 0.004 0.001 0.001 0.001 0.001 0.001
NA4  0.914 0.666 0.153 0.015 0.006 0.004 0.003 0.003
CA16 0.691 0.610 0.460 0.186 0.033 0.013 0.006 0.004
CAlc 0.562 0.563 0.564 0.564 0.564 0.565 0.565 0.565
-- per-look Pd at SCR 10 dB, Doppler opposite (looks j=0..7) --
IN1  0.885 0.940 0.905 0.796 0.165 0.014 0.004 0.002
G1   0.887 0.942 0.909 0.817 0.127 0.014 0.005 0.002
IN4  0.891 0.927 0.763 0.087 0.009 0.002 0.001 0.001
NA4  0.914 0.945 0.931 0.831 0.675 0.500 0.380 0.351
CA16 0.691 0.591 0.454 0.268 0.090 0.040 0.015 0.008
CAlc 0.562 0.564 0.564 0.563 0.564 0.564 0.565 0.562
-- SCR (dB) for cumulative Pd=0.5 by K, Doppler random; gain vs CAlc [95% day-cluster CI] --
IN1  K=1: 9.82 [ 6.40,12.68] | K=2: 9.92 [ 6.58,12.67] | K=4: 9.19 [ 5.99,11.87] | K=8: 8.42 [ 5.22,11.05]
NA4  K=1: 10.93 [ 7.46,13.74] | K=2: 10.34 [ 7.31,12.81] | K=4: 9.82 [ 7.17,11.99] | K=8: 9.40 [ 7.19,11.28]
IN4  K=1: 10.23 [ 6.71,13.18] | K=2: 9.88 [ 6.44,12.62] | K=4: 9.14 [ 5.82,11.82] | K=8: 8.28 [ 4.83,11.05]
G1   K=1: 9.90 [ 6.58,12.74] | K=2: 9.97 [ 6.75,12.63] | K=4: 9.27 [ 6.23,11.85] | K=8: 8.53 [ 5.56,11.06]
CA16 K=1: 2.76 [ 1.81, 4.06] | K=2: 2.94 [ 1.88, 4.41] | K=4: 2.85 [ 1.67, 4.49] | K=8: 2.45 [ 1.06, 4.25]
-- SCR (dB) for cumulative Pd=0.9 by K, Doppler random; gain vs CAlc [95% day-cluster CI] --
IN1  K=1: 8.95 [ 5.56,11.69] | K=2: 10.00 [ 6.21,12.95] | K=4: 9.96 [ 5.96,13.07] | K=8: 9.45 [ 5.43,12.73]
NA4  K=1: 11.09 [ 7.22,14.10] | K=2: 12.33 [ 7.84,15.68] | K=4: 12.94 [ 8.38,16.44] | K=8: 13.26 [ 8.80,16.82]
IN4  K=1: 9.65 [ 5.85,12.64] | K=2: 10.99 [ 6.43,14.37] | K=4: 10.81 [ 6.09,14.37] | K=8: 10.15 [ 5.18,13.94]
G1   K=1: 9.01 [ 5.66,11.73] | K=2: 10.07 [ 6.39,13.05] | K=4: 10.05 [ 6.23,13.22] | K=8: 9.56 [ 5.56,12.83]
CA16 K=1: 1.41 [ 0.86, 2.14] | K=2: 1.43 [ 0.76, 2.32] | K=4: 1.20 [ 0.38, 2.32] | K=8: 0.53 [-0.47, 1.81]

=== per-look Pfa design 0.001 ===
clean cumulative false-alarm prob, K=1,2,4,8: IN1      0.0011 0.0017 0.0033 0.0063
clean cumulative false-alarm prob, K=1,2,4,8: G1       0.0013 0.0021 0.0040 0.0079
clean cumulative false-alarm prob, K=1,2,4,8: IN4       0.0008 0.0014 0.0027 0.0055
clean cumulative false-alarm prob, K=1,2,4,8: NA4       0.0007 0.0014 0.0027 0.0051
clean cumulative false-alarm prob, K=1,2,4,8: CA16      0.0007 0.0015 0.0025 0.0051
clean cumulative false-alarm prob, K=1,2,4,8: CAlc      0.0010 0.0012 0.0018 0.0029
-- per-look Pd at SCR 10 dB, Doppler random (looks j=0..7) --
IN1  0.804 0.587 0.347 0.025 0.001 0.000 0.000 0.000
G1   0.833 0.626 0.455 0.014 0.002 0.000 0.000 0.000
IN4  0.802 0.559 0.133 0.010 0.000 0.000 0.000 0.000
NA4  0.852 0.790 0.516 0.312 0.173 0.124 0.114 0.110
CA16 0.555 0.368 0.125 0.025 0.006 0.002 0.001 0.000
CAlc 0.290 0.290 0.289 0.290 0.289 0.291 0.289 0.289
-- per-look Pd at SCR 10 dB, Doppler matched (looks j=0..7) --
IN1  0.804 0.003 0.000 0.000 0.000 0.000 0.000 0.000
G1   0.833 0.008 0.002 0.001 0.000 0.000 0.000 0.000
IN4  0.802 0.002 0.000 0.000 0.000 0.000 0.000 0.000
NA4  0.852 0.448 0.023 0.001 0.000 0.000 0.000 0.000
CA16 0.555 0.364 0.066 0.004 0.001 0.000 0.000 0.000
CAlc 0.290 0.289 0.291 0.290 0.289 0.289 0.290 0.290
-- per-look Pd at SCR 10 dB, Doppler opposite (looks j=0..7) --
IN1  0.804 0.878 0.629 0.057 0.001 0.000 0.000 0.000
G1   0.833 0.904 0.801 0.025 0.002 0.001 0.000 0.000
IN4  0.802 0.829 0.278 0.001 0.000 0.000 0.000 0.000
NA4  0.852 0.906 0.849 0.540 0.337 0.273 0.259 0.252
CA16 0.555 0.372 0.117 0.032 0.004 0.002 0.000 0.000
CAlc 0.290 0.290 0.291 0.290 0.289 0.292 0.290 0.291
-- SCR (dB) for cumulative Pd=0.5 by K, Doppler random; gain vs CAlc [95% day-cluster CI] --
IN1  K=1: 10.93 [ 7.28,14.02] | K=2: 11.99 [ 7.93,15.35] | K=4: 11.48 [ 7.26,15.09] | K=8: 10.88 [ 6.53,14.62]
NA4  K=1: 12.40 [ 9.09,15.50] | K=2: 13.19 [ 9.42,16.38] | K=4: 12.98 [ 9.27,15.90] | K=8: 12.73 [ 9.29,15.59]
IN4  K=1: 11.17 [ 7.43,14.42] | K=2: 12.12 [ 7.77,15.88] | K=4: 11.49 [ 7.08,15.34] | K=8: 10.87 [ 6.35,14.85]
G1   K=1: 11.63 [ 8.41,14.50] | K=2: 12.89 [ 9.11,16.16] | K=4: 12.54 [ 8.65,15.96] | K=8: 11.97 [ 8.03,15.55]
CA16 K=1: 4.06 [ 2.75, 5.93] | K=2: 4.00 [ 2.44, 6.10] | K=4: 3.57 [ 1.85, 5.85] | K=8: 3.00 [ 1.14, 5.43]
-- SCR (dB) for cumulative Pd=0.9 by K, Doppler random; gain vs CAlc [95% day-cluster CI] --
IN1  K=1: 9.54 [ 5.98,12.63] | K=2: 10.64 [ 6.86,13.92] | K=4: 10.38 [ 6.25,13.85] | K=8: 9.80 [ 5.45,13.28]
NA4  K=1: 11.30 [ 7.70,14.40] | K=2: 12.63 [ 8.56,16.01] | K=4: 13.09 [ 9.01,16.52] | K=8: 13.06 [ 9.04,16.55]
IN4  K=1: 9.86 [ 5.87,13.40] | K=2: 11.02 [ 6.57,14.94] | K=4: 10.61 [ 5.84,14.63] | K=8: 10.00 [ 5.16,14.13]
G1   K=1: 10.17 [ 6.99,13.18] | K=2: 11.42 [ 7.90,14.55] | K=4: 11.26 [ 7.42,14.46] | K=8: 10.68 [ 6.87,14.03]
CA16 K=1: 2.45 [ 1.56, 3.81] | K=2: 2.39 [ 1.22, 4.07] | K=4: 2.00 [ 0.67, 3.77] | K=8: 1.40 [-0.11, 3.33]

```

Exploratory: comparison at equal dwell false-alarm probability (0.01), with max and sum-of-squares dwell statistics:

EXPLORATORY: units 56; dwell Pfa design 0.01; persistent Swerling-1 targets, random Doppler

```

-- K = 1 looks --
IN1  max: measured dwell Pfa 0.0099 | SCR for Pd 0.5 -0.9 dB | Pd 0.9 8.8 dB
IN1  sum: measured dwell Pfa 0.0099 | SCR for Pd 0.5 -0.9 dB | Pd 0.9 8.8 dB
NA4  max: measured dwell Pfa 0.0095 | SCR for Pd 0.5 -1.9 dB | Pd 0.9 6.6 dB
NA4  sum: measured dwell Pfa 0.0095 | SCR for Pd 0.5 -1.9 dB | Pd 0.9 6.6 dB
CAlc max: measured dwell Pfa 0.0108 | SCR for Pd 0.5 9.0 dB | Pd 0.9 17.8 dB
CAlc sum: measured dwell Pfa 0.0108 | SCR for Pd 0.5 9.0 dB | Pd 0.9 17.8 dB
gain of IN1-max over integrated power CFAR (CAlc-sum) at Pd 0.5: 9.93 [ 6.46, 12.81] dB
gain of IN1-max over integrated power CFAR (CAlc-sum) at Pd 0.9: 8.94 [ 5.54, 11.59] dB
gain of NA4-max over integrated power CFAR (CAlc-sum) at Pd 0.5: 10.99 [ 7.58, 13.81] dB
gain of NA4-max over integrated power CFAR (CAlc-sum) at Pd 0.9: 11.20 [ 7.46, 14.19] dB
gain of IN1-sum over integrated power CFAR (CAlc-sum) at Pd 0.5: 9.93 [ 6.37, 12.81] dB
gain of IN1-sum over integrated power CFAR (CAlc-sum) at Pd 0.9: 8.94 [ 5.54, 11.59] dB
gain of NA4-sum over integrated power CFAR (CAlc-sum) at Pd 0.5: 10.99 [ 7.61, 13.81] dB
gain of NA4-sum over integrated power CFAR (CAlc-sum) at Pd 0.9: 11.20 [ 7.46, 14.19] dB

-- K = 2 looks --
IN1  max: measured dwell Pfa 0.0102 | SCR for Pd 0.5 -2.0 dB | Pd 0.9 7.1 dB
IN1  sum: measured dwell Pfa 0.0097 | SCR for Pd 0.5 -2.2 dB | Pd 0.9 6.8 dB
NA4  max: measured dwell Pfa 0.0093 | SCR for Pd 0.5 -2.5 dB | Pd 0.9 4.8 dB
NA4  sum: measured dwell Pfa 0.0094 | SCR for Pd 0.5 -2.7 dB | Pd 0.9 4.1 dB
CAlc max: measured dwell Pfa 0.0110 | SCR for Pd 0.5 8.1 dB | Pd 0.9 16.8 dB
CAlc sum: measured dwell Pfa 0.0108 | SCR for Pd 0.5 8.7 dB | Pd 0.9 17.2 dB
gain of IN1-max over integrated power CFAR (CAlc-sum) at Pd 0.5: 10.72 [ 6.85, 13.68] dB
gain of IN1-max over integrated power CFAR (CAlc-sum) at Pd 0.9: 10.12 [ 6.29, 13.09] dB
gain of NA4-max over integrated power CFAR (CAlc-sum) at Pd 0.5: 11.15 [ 7.71, 13.85] dB
gain of NA4-max over integrated power CFAR (CAlc-sum) at Pd 0.9: 12.46 [ 7.90, 15.85] dB
gain of IN1-sum over integrated power CFAR (CAlc-sum) at Pd 0.5: 10.91 [ 7.32, 13.74] dB
gain of IN1-sum over integrated power CFAR (CAlc-sum) at Pd 0.9: 10.40 [ 6.73, 13.29] dB
gain of NA4-sum over integrated power CFAR (CAlc-sum) at Pd 0.5: 11.42 [ 8.19, 13.92] dB
gain of NA4-sum over integrated power CFAR (CAlc-sum) at Pd 0.9: 13.13 [ 8.71, 16.43] dB

```

```
-- K = 4 looks --
IN1 max: measured dwell Pfa 0.0094 | SCR for Pd 0.5 -1.9 dB | Pd 0.9 7.4 dB
IN1 sum: measured dwell Pfa 0.0099 | SCR for Pd 0.5 -2.6 dB | Pd 0.9 6.6 dB
NA4 max: measured dwell Pfa 0.0087 | SCR for Pd 0.5 -2.7 dB | Pd 0.9 4.1 dB
NA4 sum: measured dwell Pfa 0.0099 | SCR for Pd 0.5 -3.3 dB | Pd 0.9 2.7 dB
CAloc max: measured dwell Pfa 0.0109 | SCR for Pd 0.5 7.6 dB | Pd 0.9 16.2 dB
CAloc sum: measured dwell Pfa 0.0111 | SCR for Pd 0.5 8.3 dB | Pd 0.9 16.8 dB
gain of IN1-max over integrated power CFAR (CAloc-sum) at Pd 0.5: 10.29 [ 6.26, 13.55] dB
gain of IN1-max over integrated power CFAR (CAloc-sum) at Pd 0.9: 9.47 [ 5.33, 12.64] dB
gain of NA4-max over integrated power CFAR (CAloc-sum) at Pd 0.5: 11.08 [ 7.57, 13.82] dB
gain of NA4-max over integrated power CFAR (CAloc-sum) at Pd 0.9: 12.70 [ 7.98, 16.26] dB
gain of IN1-sum over integrated power CFAR (CAloc-sum) at Pd 0.5: 10.94 [ 7.48, 13.70] dB
gain of IN1-sum over integrated power CFAR (CAloc-sum) at Pd 0.9: 10.22 [ 6.66, 13.08] dB
gain of NA4-sum over integrated power CFAR (CAloc-sum) at Pd 0.5: 11.67 [ 8.74, 13.95] dB
gain of NA4-sum over integrated power CFAR (CAloc-sum) at Pd 0.9: 14.17 [ 9.70, 17.61] dB

-- K = 8 looks --
IN1 max: measured dwell Pfa 0.0093 | SCR for Pd 0.5 -1.3 dB | Pd 0.9 8.3 dB
IN1 sum: measured dwell Pfa 0.0102 | SCR for Pd 0.5 -2.7 dB | Pd 0.9 7.0 dB
NA4 max: measured dwell Pfa 0.0091 | SCR for Pd 0.5 -2.9 dB | Pd 0.9 4.3 dB
NA4 sum: measured dwell Pfa 0.0100 | SCR for Pd 0.5 -3.8 dB | Pd 0.9 2.4 dB
CAloc max: measured dwell Pfa 0.0103 | SCR for Pd 0.5 7.4 dB | Pd 0.9 16.0 dB
CAloc sum: measured dwell Pfa 0.0105 | SCR for Pd 0.5 8.0 dB | Pd 0.9 16.4 dB
gain of IN1-max over integrated power CFAR (CAloc-sum) at Pd 0.5: 9.31 [ 4.87, 12.82] dB
gain of IN1-max over integrated power CFAR (CAloc-sum) at Pd 0.9: 8.14 [ 3.83, 11.48] dB
gain of NA4-max over integrated power CFAR (CAloc-sum) at Pd 0.5: 10.85 [ 7.29, 13.54] dB
gain of NA4-max over integrated power CFAR (CAloc-sum) at Pd 0.9: 12.13 [ 7.50, 15.64] dB
gain of IN1-sum over integrated power CFAR (CAloc-sum) at Pd 0.5: 10.70 [ 7.35, 13.40] dB
gain of IN1-sum over integrated power CFAR (CAloc-sum) at Pd 0.9: 9.35 [ 6.02, 12.16] dB
gain of NA4-sum over integrated power CFAR (CAloc-sum) at Pd 0.5: 11.74 [ 9.17, 13.75] dB
gain of NA4-sum over integrated power CFAR (CAloc-sum) at Pd 0.9: 14.01 [ 9.97, 17.21] dB
```

Exploratory: exact-law prediction of the post-onset decay:

```
Exact-law prediction vs measured per-look Pd after onset (IN-ARCP, SCR 10 dB, Pfa 0.01); pooled over 56 units (exploratory)
random predicted 0.885 0.710 0.635 0.472 0.086 0.012 0.004 0.002
measured 0.885 0.711 0.639 0.478 0.084 0.012 0.005 0.002 MAE per unit x look 0.010
matched predicted 0.885 0.023 0.013 0.007 0.004 0.002 0.002 0.001
measured 0.885 0.024 0.013 0.007 0.003 0.002 0.002 0.001 MAE per unit x look 0.003
opposite predicted 0.885 0.941 0.906 0.793 0.167 0.014 0.004 0.002
measured 0.885 0.940 0.905 0.796 0.165 0.014 0.004 0.002 MAE per unit x look 0.008
```

10 Numerical checks of the detection theory

Monte Carlo checks of Corollary 2, Proposition 2 (closed forms) and Proposition 3 (output of `detection/theory_checks.py`):

```
T1/T2 exact detection law vs Monte Carlo (Swirling-1 target in Y only; 400k draws)
rho=0.91-0.18j c=None r=0.91-0.18j SCR=1.0: MC 0.5358 exact 0.5364 closed form 0.5364
rho=0.91-0.18j c=10.0 r=0.88-0.18j SCR=3.0: MC 0.6381 exact 0.6374
rho=0.70 c=1.0 r=0.65 SCR=10.0: MC 0.4275 exact 0.4263

T3 noise-limited whitening gain: empirical (c+1)/E[Y-pred]^2 vs closed forms (m=16 history, 200k draws)
|rho|=0.7 c= 0.1: IN emp -1.46 dB closed -1.46 | oracle emp 0.03 dB closed(m=inf) 0.03 | clutter-only bound 2.92
|rho|=0.7 c= 1.0: IN emp -0.01 dB closed 0.00 | oracle emp 0.66 dB closed(m=inf) 0.67 | clutter-only bound 2.92
|rho|=0.7 c= 10.0: IN emp 2.24 dB closed 2.23 | oracle emp 2.29 dB closed(m=inf) 2.27 | clutter-only bound 2.92
|rho|=0.7 c= 100.0: IN emp 2.85 dB closed 2.84 | oracle emp 2.85 dB closed(m=inf) 2.84 | clutter-only bound 2.92
|rho|=0.93 c= 0.1: IN emp -2.32 dB closed -2.32 | oracle emp 0.12 dB closed(m=inf) 0.12 | clutter-only bound 8.69
|rho|=0.93 c= 1.0: IN emp 0.02 dB closed 0.00 | oracle emp 1.66 dB closed(m=inf) 1.65 | clutter-only bound 8.69
|rho|=0.93 c= 10.0: IN emp 5.34 dB closed 5.34 | oracle emp 5.76 dB closed(m=inf) 5.76 | clutter-only bound 8.69
|rho|=0.93 c= 100.0: IN emp 8.17 dB closed 8.18 | oracle emp 8.18 dB closed(m=inf) 8.19 | clutter-only bound 8.69
|rho|=0.98 c= 0.1: IN emp -2.50 dB closed -2.52 | oracle emp 0.20 dB closed(m=inf) 0.21 | clutter-only bound 14.02
|rho|=0.98 c= 1.0: IN emp -0.01 dB closed 0.00 | oracle emp 2.22 dB closed(m=inf) 2.22 | clutter-only bound 14.02
|rho|=0.98 c= 10.0: IN emp 6.69 dB closed 6.69 | oracle emp 7.78 dB closed(m=inf) 7.78 | clutter-only bound 14.02
|rho|=0.98 c= 100.0: IN emp 12.31 dB closed 12.32 | oracle emp 12.43 dB closed(m=inf) 12.44 | clutter-only bound 14.02

T4 persistent strong target: limiting score and Pd (IN-ARCP, r fitted = 0.93e*(-0.2j))
sqrt(m/(m-1)) = 1.033; thresholds q(0.1)=1.574 q(0.01)=2.310
Doppler w=+0.0: limiting score 0.937 | SCR 0 dB: Pd(0.01) 0.010 | SCR 20 dB: Pd(0.01) 0.000 | SCR 40 dB: Pd(0.01) 0.000
Doppler w=-0.2: limiting score 0.613 | SCR 0 dB: Pd(0.01) 0.010 | SCR 20 dB: Pd(0.01) 0.000 | SCR 40 dB: Pd(0.01) 0.000
Doppler w=+1.0: limiting score 1.029 | SCR 0 dB: Pd(0.01) 0.000 | SCR 20 dB: Pd(0.01) 0.000 | SCR 40 dB: Pd(0.01) 0.000
```

11 Baselines: parametric texture and adaptive conformal inference

B1/B2 within-session units: 56; transfer days: 6

```
=== 1-alpha = 0.9 ===
B1 NA4-K / NA4 radius (GM): 1.0092 [1.0064, 1.0115] | coverage NA4 0.9019 NA4-K 0.9015 | max dev NA4 0.043 NA4-K 0.043
B2 within ACI g=0.005: coverage 0.9010 vs split 0.9013 | radius ratio (finite) 0.999 [0.997,1.000] | infinite-region fraction 0.0000 | max dev 0.033 vs 0.036
B2 within ACI g=0.02: coverage 0.9007 vs split 0.9013 | radius ratio (finite) 0.999 [0.995,1.002] | infinite-region fraction 0.0000 | max dev 0.029 vs 0.036
B2 transfer IN1 : mean coverage 0.9029 worst day 0.8822 | radius vs split 1.000 | max dev 0.024 | per day 0.905 0.894 0.915 0.918 0.904 0.882
B2 transfer ACIO.005 : mean coverage 0.9032 worst day 0.8889 | radius vs split 1.001 | max dev 0.020 | per day 0.904 0.897 0.912 0.914 0.903 0.889
B2 transfer ACIO.02 : mean coverage 0.9024 worst day 0.8965 | radius vs split 1.000 | max dev 0.015 | per day 0.903 0.900 0.906 0.908 0.902 0.897

=== 1-alpha = 0.99 ===
B1 NA4-K / NA4 radius (GM): 1.0078 [1.0034, 1.0104] | coverage NA4 0.9905 NA4-K 0.9906 | max dev NA4 0.011 NA4-K 0.011
B2 within ACI g=0.005: coverage 0.9903 vs split 0.9901 | radius ratio (finite) 1.001 [0.996,1.007] | infinite-region fraction 0.0000 | max dev 0.010 vs 0.011
B2 within ACI g=0.02: coverage 0.9901 vs split 0.9901 | radius ratio (finite) 1.006 [0.998,1.015] | infinite-region fraction 0.0115 | max dev 0.009 vs 0.011
B2 transfer IN1 : mean coverage 0.9921 worst day 0.9875 | radius vs split 1.000 | max dev 0.006 | per day 0.987 0.992 0.994 0.996 0.994 0.989
B2 transfer ACIO.005 : mean coverage 0.9918 worst day 0.9889 | radius vs split 0.997 | max dev 0.005 | per day 0.989 0.991 0.993 0.995 0.993 0.990
B2 transfer ACIO.02 : mean coverage 0.9912 worst day 0.9902 | radius vs split 0.995 | max dev 0.004 | per day 0.990 0.990 0.992 0.993 0.991 0.990
```

12 Classical detectors, gradual emergence and ACI contamination (R10)

Frozen protocol `r10/PROTOCOL_R10.md`; stated expectations and verdicts in `r10/RESULTS_R10.md`. Output of `r10/analyze_r10.py`:

units 56; days 6; test segments 74980

```
=== Part C: dwell detectors at equal dwell Pfa (alpha = 0.01 unless stated) ===
-- measured dwell Pfa on clean test segments, alpha = 0.01 (K' = 1 2 4 8) --
IN1sum 0.0099 0.0097 0.0099 0.0102
NA4sum 0.0095 0.0094 0.0099 0.0100
```

```

CALocsum 0.0108 0.0108 0.0111 0.0105
PAMFH 0.0098 0.0109 0.0101 0.0096
NPAMF 0.0098 0.0109 0.0098 0.0101
MTDhann nan 0.0095 0.0095 0.0106
MTDrect nan 0.0095 0.0096 0.0097
-- measured dwell Pfa on clean test segments, alpha = 0.001 (K' = 1 2 4 8) --
IN1sum 0.0011 0.0008 0.0008 0.0009
NA4sum 0.0007 0.0011 0.0009 0.0010
CALocsum 0.0009 0.0010 0.0011 0.0012
PAMFH 0.0008 0.0010 0.0009 0.0010
NPAMF 0.0008 0.0009 0.0009 0.0012
MTDhann nan 0.0005 0.0009 0.0008
MTDrect nan 0.0005 0.0010 0.0009
-- SCR (dB) for dwell Pd = 0.5 / 0.9, Doppler random, alpha 0.01; mean over units --
IN1sum -0.8/ 8.7 -2.2/ 6.7 -2.5/ 6.5 -2.7/ 6.9
NA4sum -1.9/ 6.6 -2.7/ 4.1 -3.3/ 2.7 -3.7/ 2.3
CALocsum 9.0/ 17.7 8.7/ 17.2 8.4/ 16.8 8.0/ 16.4
PAMFH -1.3/ 8.0 -2.4/ 5.1 -3.1/ 3.7 -3.7/ 3.0
NPAMF -1.3/ 8.0 -2.4/ 5.1 -3.1/ 3.6 -3.8/ 2.6
MTDhann - 8.4/ 20.7 4.5/ 19.2 2.3/ 17.9
MTDrect - 8.4/ 20.7 7.7/ 19.9 6.3/ 18.9
-- SCR (dB) for dwell Pd = 0.5 / 0.9, Doppler random, alpha 0.001; mean over units --
IN1sum 2.0/ 11.5 0.3/ 10.0 0.4/ 10.3 1.0/ 11.3
NA4sum 0.5/ 9.8 -1.3/ 7.5 -1.3/ 6.8 -1.4/ 7.2
CALocsum 13.1/ 21.0 12.8/ 20.5 12.6/ 20.2 12.2/ 20.1
PAMFH 1.8/ 11.2 -0.4/ 8.9 -1.2/ 7.6 -2.0/ 7.0
NPAMF 1.8/ 11.2 -0.6/ 8.6 -2.0/ 6.4 -3.0/ 5.0
MTDhann - 14.8/ 22.4 11.9/ 22.5 10.6/ 22.2
MTDrect - 14.8/ 22.4 14.9/ 22.6 13.8/ 22.1
-- SCR (dB) for dwell Pd = 0.5 / 0.9, Doppler matched, alpha 0.01; mean over units --
IN1sum -0.8/ 8.7 0.3/ 10.0 2.1/ 11.6 4.0/ 13.4
NA4sum -1.9/ 6.6 -2.0/ 6.7 -1.5/ 7.5 -0.1/ 9.2
CALocsum 9.0/ 17.7 8.8/ 17.5 8.5/ 17.2 8.2/ 16.8
PAMFH -1.3/ 8.0 -1.0/ 8.4 -0.3/ 9.1 0.6/ 10.1
NPAMF -1.3/ 8.0 -1.0/ 8.4 -0.3/ 9.0 0.2/ 9.5
MTDhann - 18.3/ 22.1 18.3/ 21.6 19.0/ 20.2
MTDrect - 18.3/ 22.1 18.3/ 21.4 17.8/ 21.7
-- SCR (dB) for dwell Pd = 0.5 / 0.9, Doppler opposite, alpha 0.01; mean over units --
IN1sum -0.8/ 8.7 -3.0/ 3.6 -3.6/ 1.9 -3.8/ 1.3
NA4sum -1.9/ 6.6 -3.1/ 1.9 -3.7/ 0.3 -4.1/ -0.8
CALocsum 9.0/ 17.7 8.5/ 16.9 8.3/ 16.6 7.9/ 16.3
PAMFH -1.3/ 8.0 -3.0/ 2.2 -3.6/ 0.1 -4.1/ -1.1
NPAMF -1.3/ 8.0 -3.0/ 2.2 -3.6/ -0.1 -4.2/ -1.3
MTDhann - 4.5/ 15.1 0.3/ 9.4 -0.5/ 8.5
MTDrect - 4.5/ 15.1 4.0/ 14.6 2.6/ 13.0
-- gains (dB, positive = first detector needs less SCR) at Pd = 0.5, random Doppler, alpha 0.01; 95% day bootstrap --
K'=1 Pd=0.5 IN1sum vs PAMFH : -0.43 [ -0.59, -0.24] LOD0 -0.48..-0.37
K'=1 Pd=0.5 IN1sum vs PAMFH : -0.72 [ -1.07, -0.26] LOD0 -0.86..-0.44
K'=1 Pd=0.5 NA4sum vs PAMFH : 0.66 [ 0.44, 0.91] LOD0 0.59.. 0.75
K'=1 Pd=0.5 NA4sum vs PAMFH : 1.43 [ 1.37, 1.51] LOD0 1.41.. 1.45
K'=1 Pd=0.5 IN1sum vs CALocsum: 9.86 [ 6.31, 12.74] LOD0 8.99..11.60
K'=1 Pd=0.5 IN1sum vs CALocsum: 9.00 [ 5.58, 11.67] LOD0 8.07..10.65
K'=1 Pd=0.5 PAMFH vs CALocsum: 10.29 [ 6.79, 13.25] LOD0 9.39..12.07
K'=1 Pd=0.5 PAMFH vs CALocsum: 9.72 [ 6.08, 12.63] LOD0 8.51..11.52
K'=1 Pd=0.5 PAMFH vs NPAMF : 0.00 [ 0.00, 0.00] LOD0 0.00.. 0.00
K'=1 Pd=0.5 PAMFH vs NPAMF : 0.00 [ 0.00, 0.00] LOD0 0.00.. 0.00
K'=2 Pd=0.5 IN1sum vs PAMFH : -0.20 [ -0.30, 0.00] LOD0 -0.24..-0.15
K'=2 Pd=0.5 IN1sum vs PAMFH : -1.68 [ -2.39, -0.77] LOD0 -1.99..-1.10
K'=2 Pd=0.5 NA4sum vs PAMFH : 0.32 [ 0.07, 0.67] LOD0 0.17.. 0.47
K'=2 Pd=0.5 NA4sum vs PAMFH : 1.00 [ 0.84, 1.19] LOD0 0.93.. 1.13
K'=2 Pd=0.5 IN1sum vs MTDhann: 10.59 [ 9.46, 12.27] LOD0 10.19..11.13
K'=2 Pd=0.5 IN1sum vs MTDhann: 14.19 [ 12.11, 16.48] LOD0 13.70..15.24
K'=2 Pd=0.5 NA4sum vs MTDhann: 11.11 [ 10.34, 12.46] LOD0 10.81..11.45
K'=2 Pd=0.5 NA4sum vs MTDhann: 16.88 [ 14.09, 19.13] LOD0 16.31..18.18
K'=2 Pd=0.5 PAMFH vs MTDhann: 10.79 [ 9.75, 12.25] LOD0 10.43..11.28
K'=2 Pd=0.5 PAMFH vs MTDhann: 15.92 [ 12.95, 18.05] LOD0 15.25..17.25
K'=2 Pd=0.5 IN1sum vs CALocsum: 10.84 [ 7.30, 13.75] LOD0 9.98..12.63
K'=2 Pd=0.5 IN1sum vs CALocsum: 10.48 [ 6.68, 13.38] LOD0 9.42..12.33
K'=2 Pd=0.5 PAMFH vs CALocsum: 11.04 [ 7.55, 13.80] LOD0 10.13..12.79
K'=2 Pd=0.5 PAMFH vs CALocsum: 12.16 [ 7.57, 15.49] LOD0 10.53..14.32
K'=2 Pd=0.5 MTDhann vs CALocsum: 0.26 [ -2.49, 2.31] LOD0 -0.84.. 1.50
K'=2 Pd=0.5 MTDhann vs CALocsum: -3.47 [ -5.49, -2.08] LOD0 -4.46..-2.73
K'=2 Pd=0.5 PAMFH vs NPAMF : 0.00 [ -0.00, 0.00] LOD0 0.00.. 0.00
K'=2 Pd=0.5 PAMFH vs NPAMF : 0.01 [ -0.03, 0.04] LOD0 -0.00.. 0.02
K'=4 Pd=0.5 IN1sum vs PAMFH : -0.54 [ -0.88, -0.19] LOD0 -0.63..-0.37
K'=4 Pd=0.5 IN1sum vs PAMFH : -2.78 [ -3.49, -1.87] LOD0 -3.11..-2.23
K'=4 Pd=0.5 NA4sum vs PAMFH : 0.22 [ 0.04, 0.47] LOD0 0.10.. 0.29
K'=4 Pd=0.5 NA4sum vs PAMFH : 1.00 [ 0.81, 1.27] LOD0 0.91.. 1.11
K'=4 Pd=0.5 IN1sum vs MTDhann: 6.98 [ 5.95, 8.43] LOD0 6.74.. 7.86
K'=4 Pd=0.5 IN1sum vs MTDhann: 12.84 [ 11.08, 14.97] LOD0 12.34..13.68
K'=4 Pd=0.5 NA4sum vs MTDhann: 7.74 [ 6.53, 9.23] LOD0 7.35.. 8.73
K'=4 Pd=0.5 NA4sum vs MTDhann: 16.63 [ 14.09, 18.91] LOD0 15.98..17.77
K'=4 Pd=0.5 PAMFH vs MTDhann: 7.52 [ 6.45, 8.90] LOD0 7.25.. 8.45
K'=4 Pd=0.5 PAMFH vs MTDhann: 15.68 [ 13.18, 17.78] LOD0 15.12..16.79
K'=4 Pd=0.5 IN1sum vs CALocsum: 10.88 [ 7.44, 13.64] LOD0 10.04..12.60
K'=4 Pd=0.5 IN1sum vs CALocsum: 10.32 [ 6.71, 13.19] LOD0 9.41..12.08
K'=4 Pd=0.5 PAMFH vs CALocsum: 11.42 [ 8.29, 13.88] LOD0 10.63..12.98
K'=4 Pd=0.5 PAMFH vs CALocsum: 13.10 [ 8.63, 16.42] LOD0 11.64..15.19
K'=4 Pd=0.5 MTDhann vs CALocsum: 3.90 [ -0.24, 6.88] LOD0 2.19.. 5.72
K'=4 Pd=0.5 MTDhann vs CALocsum: -2.39 [ -4.54, -0.89] LOD0 -3.36..-1.60
K'=4 Pd=0.5 PAMFH vs NPAMF : -0.04 [ -0.11, 0.01] LOD0 -0.06..-0.00
K'=4 Pd=0.5 PAMFH vs NPAMF : -0.13 [ -0.21, -0.08] LOD0 -0.16..-0.11
K'=8 Pd=0.5 IN1sum vs PAMFH : -1.01 [ -1.67, -0.34] LOD0 -1.18..-0.70
K'=8 Pd=0.5 IN1sum vs PAMFH : -3.95 [ -4.43, -3.32] LOD0 -4.17..-3.57
K'=8 Pd=0.5 NA4sum vs PAMFH : 0.08 [ -0.14, 0.30] LOD0 -0.04.. 0.14
K'=8 Pd=0.5 NA4sum vs PAMFH : 0.70 [ 0.49, 1.03] LOD0 0.62.. 0.77
K'=8 Pd=0.5 IN1sum vs MTDhann: 4.95 [ 4.19, 5.97] LOD0 4.75.. 5.58
K'=8 Pd=0.5 IN1sum vs MTDhann: 10.96 [ 9.11, 12.61] LOD0 10.54..11.83
K'=8 Pd=0.5 NA4sum vs MTDhann: 6.04 [ 4.85, 7.58] LOD0 5.41.. 6.80
K'=8 Pd=0.5 NA4sum vs MTDhann: 15.60 [ 13.19, 17.56] LOD0 14.90..16.71
K'=8 Pd=0.5 PAMFH vs MTDhann: 5.96 [ 4.85, 7.48] LOD0 5.45.. 6.71
K'=8 Pd=0.5 PAMFH vs MTDhann: 14.90 [ 12.56, 16.68] LOD0 14.14..15.99
K'=8 Pd=0.5 IN1sum vs CALocsum: 10.65 [ 7.32, 13.37] LOD0 9.82..12.34
K'=8 Pd=0.5 IN1sum vs CALocsum: 9.41 [ 5.89, 12.28] LOD0 8.54..11.12
K'=8 Pd=0.5 PAMFH vs CALocsum: 11.66 [ 8.91, 13.74] LOD0 10.94..13.03
K'=8 Pd=0.5 PAMFH vs CALocsum: 13.36 [ 9.35, 16.42] LOD0 12.11..15.29
K'=8 Pd=0.5 MTDhann vs CALocsum: 5.70 [ 1.71, 8.63] LOD0 4.24.. 7.58
K'=8 Pd=0.5 MTDhann vs CALocsum: -1.54 [ -3.31, -0.10] LOD0 -2.03..-0.71
K'=8 Pd=0.5 PAMFH vs NPAMF : -0.15 [ -0.27, -0.04] LOD0 -0.18..-0.09
K'=8 Pd=0.5 PAMFH vs NPAMF : -0.44 [ -0.55, -0.30] LOD0 -0.48..-0.40

```

```

C1 check: max |Pd(PAMF-H, K'=1) - Pd(IN4 look 0)| over SCR = 0.0000 (expectation < 0.005)
C2: PAMF-H needs 1.01 dB less SCR than IN1-sum at K'=8 (expectation >= 3 dB): FAILED
C3: MTD-CA needs 5.70 dB less SCR than Caloc-sum at K'=8 (expectation >= 5 dB): OK
C4: PAMF-H matched-Doppler SCR is 4.25 dB worse than random (expectation >= 3 dB): OK

```

```

=== Part R: gradual emergence (random Doppler, alpha 0.01) ===

```

```

-- placement pre: per-look Pd at SCR 10 dB (looks 0..7), mean over units --
L= 1 IN1      0.883 0.711 0.637 0.477 0.086 0.012 0.004 0.002
L= 1 NA4      0.914 0.883 0.702 0.573 0.433 0.290 0.205 0.183
L= 1 CALoc    0.558 0.557 0.558 0.559 0.557 0.558 0.558 0.557
L= 4 IN1      0.793 0.705 0.549 0.099 0.015 0.006 0.003 0.002
L= 4 NA4      0.790 0.821 0.643 0.473 0.281 0.203 0.183 0.173
L= 4 CALoc    0.559 0.556 0.557 0.557 0.559 0.558 0.559 0.557
L= 8 IN1      0.656 0.446 0.047 0.012 0.005 0.003 0.002 0.001
L= 8 NA4      0.678 0.684 0.404 0.237 0.194 0.179 0.171 0.166
L= 8 CALoc    0.557 0.559 0.558 0.556 0.558 0.557 0.558 0.557
L=16 IN1      0.014 0.007 0.003 0.002 0.001 0.001 0.001 0.001
L=16 NA4      0.204 0.203 0.184 0.175 0.168 0.165 0.162 0.160
L=16 CALoc    0.558 0.558 0.557 0.557 0.558 0.558 0.560 0.559

-- placement pre: SCR for Pd 0.5 (single look 0 | any of 8 looks) and first-detection look at 10 dB --
L= 1 IN1      look0 -0.8 | cum8 -3.1 | first look 0.12
L= 1 IN4      look0 -1.3 | cum8 -3.0 | first look 0.10
L= 1 NA4      look0 -1.9 | cum8 -4.1 | first look 0.14
L= 1 CA16     look0  6.2 | cum8  2.8 | first look 0.27
L= 1 CALoc    look0  9.0 | cum8  5.4 | first look 0.63
L= 4 IN1      look0  0.4 | cum8 -2.5 | first look 0.16
L= 4 IN4      look0  0.1 | cum8 -1.9 | first look 0.12
L= 4 NA4      look0 -1.7 | cum8 -4.0 | first look 0.22
L= 4 CA16     look0  7.7 | cum8  4.2 | first look 0.32
L= 4 CALoc    look0  9.0 | cum8  5.4 | first look 0.63
L= 8 IN1      look0  4.4 | cum8  0.3 | first look 0.21
L= 8 IN4      look0  9.3 | cum8  7.4 | first look 0.21
L= 8 NA4      look0 -0.9 | cum8 -4.0 | first look 0.27
L= 8 CA16     look0 11.7 | cum8  8.3 | first look 0.42
L= 8 CALoc    look0  9.1 | cum8  5.4 | first look 0.63
L=16 IN1      look0 nan | cum8 nan | first look 1.49
L=16 IN4      look0 nan | cum8 nan | first look 2.29
L=16 NA4      look0 18.7 | cum8 13.0 | first look 0.67
L=16 CA16     look0 nan | cum8 nan | first look 0.88
L=16 CALoc    look0  9.1 | cum8  5.4 | first look 0.63

-- placement pre: SCR for dwell Pd 0.5 at K=8 (dwell detectors) --
L= 1 IN1sum   -2.7 NA4sum -3.7 CALocsum  8.0 PAMFH -3.7 NPAMF -3.8 MTDhann  2.3 MTDirect  6.3
L= 4 IN1sum   -2.1 NA4sum -3.6 CALocsum  8.0 PAMFH -3.5 NPAMF -3.6 MTDhann  2.3 MTDirect  6.2
L= 8 IN1sum    0.3 NA4sum -3.5 CALocsum  8.0 PAMFH -3.1 NPAMF -3.5 MTDhann  2.3 MTDirect  6.3
L=16 IN1sum   nan NA4sum  9.3 CALocsum  8.0 PAMFH -1.8 NPAMF -3.0 MTDhann  2.3 MTDirect  6.2

-- placement in: per-look Pd at SCR 10 dB (looks 0..7), mean over units --
L= 1 IN1      0.883 0.711 0.637 0.477 0.086 0.012 0.004 0.002
L= 1 NA4      0.914 0.883 0.702 0.573 0.433 0.290 0.205 0.183
L= 1 CALoc    0.558 0.557 0.558 0.559 0.557 0.558 0.558 0.557
L= 4 IN1      0.465 0.676 0.757 0.794 0.703 0.547 0.097 0.015
L= 4 NA4      0.558 0.726 0.786 0.792 0.821 0.643 0.471 0.280
L= 4 CALoc    0.036 0.179 0.388 0.569 0.557 0.558 0.558 0.557
L= 8 IN1      0.212 0.443 0.555 0.617 0.651 0.670 0.673 0.656
L= 8 NA4      0.304 0.529 0.630 0.677 0.695 0.701 0.693 0.677
L= 8 CALoc    0.015 0.037 0.092 0.180 0.285 0.387 0.479 0.557
L=16 IN1      0.065 0.217 0.333 0.411 0.456 0.482 0.491 0.484
L=16 NA4      0.104 0.307 0.436 0.513 0.555 0.579 0.585 0.583
L=16 CALoc    0.012 0.016 0.023 0.036 0.059 0.091 0.133 0.179

-- placement in: SCR for Pd 0.5 (single look 0 | any of 8 looks) and first-detection look at 10 dB --
L= 1 IN1      look0 -0.8 | cum8 -3.1 | first look 0.12
L= 1 IN4      look0 -1.3 | cum8 -3.0 | first look 0.10
L= 1 NA4      look0 -1.9 | cum8 -4.1 | first look 0.14
L= 1 CA16     look0  6.2 | cum8  2.8 | first look 0.27
L= 1 CALoc    look0  9.0 | cum8  5.4 | first look 0.63
L= 4 IN1      look0 11.0 | cum8 -2.8 | first look 0.91
L= 4 IN4      look0 10.2 | cum8 -2.5 | first look 0.82
L= 4 NA4      look0  8.9 | cum8 -3.8 | first look 0.77
L= 4 CA16     look0 18.2 | cum8  3.7 | first look 1.70
L= 4 CALoc    look0 21.0 | cum8  6.0 | first look 2.46
L= 8 IN1      look0 16.2 | cum8 -2.0 | first look 1.94
L= 8 IN4      look0 15.4 | cum8 -1.6 | first look 1.73
L= 8 NA4      look0 14.3 | cum8 -3.0 | first look 1.60
L= 8 CA16     look0 23.6 | cum8  5.6 | first look 3.23
L= 8 CALoc    look0 24.8 | cum8  7.5 | first look 4.35
L=16 IN1      look0 19.9 | cum8  3.1 | first look 2.84
L=16 IN4      look0 18.9 | cum8  2.9 | first look 2.58
L=16 NA4      look0 17.7 | cum8 -0.2 | first look 2.54
L=16 CA16     look0 nan | cum8 11.6 | first look 4.01
L=16 CALoc    look0 nan | cum8 13.6 | first look 4.99

-- placement in: SCR for dwell Pd 0.5 at K=8 (dwell detectors) --
L= 1 IN1sum   -2.7 NA4sum -3.7 CALocsum  8.0 PAMFH -3.7 NPAMF -3.8 MTDhann  2.3 MTDirect  6.3
L= 4 IN1sum   -2.1 NA4sum -3.2 CALocsum  9.3 PAMFH -3.0 NPAMF -3.1 MTDhann  2.9 MTDirect  7.7
L= 8 IN1sum   -0.3 NA4sum -2.3 CALocsum 12.0 PAMFH -1.8 NPAMF -1.7 MTDhann  6.8 MTDirect 10.7
L=16 IN1sum    5.6 NA4sum  1.7 CALocsum 18.0 PAMFH  2.8 NPAMF  3.2 MTDhann 12.9 MTDirect 16.8
R-pre single-look gain IN1 vs CALoc, L= 1:  9.86 [ 6.31, 12.78] LOD0  8.99..11.60
R-pre single-look gain IN1 vs CALoc, L= 4:  8.65 [ 5.11, 11.50] LOD0  7.82..10.40
R-pre single-look gain IN1 vs CALoc, L= 8:  4.65 [ 0.90,  7.62] LOD0  3.71.. 6.54
R-pre single-look gain IN1 vs CALoc, L=16: nan [ nan, nan] LOD0 nan.. nan
R1: gain decreasing in L: False; below 5 dB at L=16: False -> FAILED
R3: IN1 first-detection look (R-in) by L: [0.12, 0.91, 1.94, 2.84] -> OK

=== Part A: ACI with target-contaminated feedback (IN1, alpha 0.01) ===
pi=0 split Pfa/a 0.99 Pd nan | naive0.005 Pfa/a 0.97 Pd nan | gated0.005 Pfa/a 0.97 Pd nan | naive0.02 Pfa/a 0.99 Pd nan | gated0.02 Pfa/a 0.99 Pd nan
pi=0.01 split Pfa/a 0.99 Pd 0.893 | naive0.005 Pfa/a 0.68 Pd 0.880 | gated0.005 Pfa/a 0.89 Pd 0.888 | naive0.02 Pfa/a 0.52 Pd 0.664 | gated0.02 Pfa/a 0.82 Pd 0.850
pi=0.05 split Pfa/a 0.99 Pd 0.890 | naive0.005 Pfa/a 0.24 Pd 0.426 | gated0.005 Pfa/a 0.68 Pd 0.834 | naive0.02 Pfa/a 0.14 Pd 0.275 | gated0.02 Pfa/a 0.55 Pd 0.635
A1 (gamma 0.02): naive clean Pfa/alpha 0.14 (< 0.87) and Pd 0.275 vs split 0.890 -> OK
A2 (gamma 0.02): gated clean Pfa/alpha 0.55 within [0.8, 1.2]? -> FAILED

```

Exact-law prediction of IN-ARCP's per-look detection probability under gradual emergence (endpoint R2):

```

pre|L1      pred 0.885 0.710 0.635 0.472 0.086 0.012 0.004 0.002
           meas 0.883 0.711 0.637 0.477 0.086 0.012 0.004 0.002 | MAE 0.009
pre|L4      pred 0.789 0.702 0.541 0.101 0.015 0.006 0.003 0.002
           meas 0.793 0.705 0.549 0.099 0.015 0.006 0.003 0.002 | MAE 0.009
pre|L8      pred 0.648 0.434 0.049 0.012 0.005 0.003 0.002 0.001
           meas 0.656 0.446 0.047 0.012 0.005 0.003 0.002 0.001 | MAE 0.008
pre|L16     pred 0.015 0.007 0.004 0.002 0.002 0.001 0.001 0.001
           meas 0.014 0.007 0.003 0.002 0.001 0.001 0.001 0.001 | MAE 0.001
in|L1      pred 0.885 0.710 0.635 0.472 0.086 0.012 0.004 0.002
           meas 0.883 0.711 0.637 0.477 0.086 0.012 0.004 0.002 | MAE 0.009
in|L4      pred 0.429 0.665 0.750 0.789 0.702 0.541 0.101 0.015
           meas 0.465 0.676 0.757 0.794 0.703 0.547 0.097 0.015 | MAE 0.021
in|L8      pred 0.155 0.407 0.535 0.604 0.643 0.662 0.665 0.648
           meas 0.212 0.443 0.555 0.617 0.651 0.670 0.673 0.656 | MAE 0.033
in|L16     pred 0.038 0.168 0.296 0.379 0.432 0.461 0.472 0.462
           meas 0.065 0.217 0.333 0.411 0.456 0.482 0.491 0.484 | MAE 0.037
R2: exact-law MAE per unit x look x L x placement = 0.016 (expectation <= 0.03): OK

```

13 Audit of existing outcomes: conditional false-alarm rates, dependence, leave-one-day-out, fit diagnostics

Output of `r10/audit_existing.py` (descriptive and sensitivity analyses requested in review; no new runs):

```
=== 1. IPIX false-alarm rate by texture quintile (confirmatory, m = 16, +/-1024 proxy) ===
-- design alpha = 0.1: Pfa/alpha per quintile (low -> high texture), mean over units; exceedances pooled --
IN1 1.04 0.98 1.00 0.99 0.92 | spread 1.1x | exceedances 1557 1466 1501 1483 1373 of 74980 episodes
IN4 1.00 0.99 0.97 1.00 0.97 | spread 1.0x | exceedances 1495 1474 1456 1501 1455 of 74980 episodes
NA4 0.84 0.96 1.06 1.02 1.02 | spread 1.3x | exceedances 1252 1436 1591 1540 1538 of 74980 episodes
U 0.22 0.44 0.71 1.12 2.56 | spread 11.5x | exceedances 325 662 1057 1680 3846 of 74980 episodes
RMS 1.88 1.25 0.88 0.60 0.43 | spread 4.4x | exceedances 2813 1876 1321 899 636 of 74980 episodes
MLP 1.82 1.17 0.84 0.61 0.50 | spread 3.6x | exceedances 2732 1761 1264 927 749 of 74980 episodes
MON 0.79 0.86 0.98 1.05 1.13 | spread 1.4x | exceedances 1177 1277 1464 1579 1689 of 74980 episodes
-- design alpha = 0.01: Pfa/alpha per quintile (low -> high texture), mean over units; exceedances pooled --
IN1 1.18 1.03 1.14 0.91 0.67 | spread 1.7x | exceedances 175 155 172 137 101 of 74980 episodes
IN4 1.09 0.99 0.93 0.99 0.93 | spread 1.2x | exceedances 163 149 141 150 140 of 74980 episodes
NA4 0.72 0.94 0.94 0.93 1.22 | spread 1.7x | exceedances 107 139 142 139 184 of 74980 episodes
U 0.05 0.18 0.21 0.60 4.68 | spread 87.5x | exceedances 8 27 32 89 710 of 74980 episodes
RMS 2.62 1.25 0.83 0.43 0.27 | spread 9.8x | exceedances 390 187 124 64 40 of 74980 episodes
MLP 2.35 1.13 0.81 0.45 0.48 | spread 5.2x | exceedances 353 167 121 67 71 of 74980 episodes
MON 0.50 0.60 0.92 0.83 1.14 | spread 2.3x | exceedances 74 90 139 125 171 of 74980 episodes
-- maximum quintile deviation: observed vs binomial reference vs dependence-aware (time-shift) reference, alpha = 0.1 --
IN1 observed 0.036 | binomial ref 0.029 | time-shift ref 0.033
IN4 observed 0.032 | binomial ref 0.029 | time-shift ref 0.033
NA4 observed 0.043 | binomial ref 0.029 | time-shift ref 0.039
U observed 0.165 | binomial ref 0.029 | time-shift ref 0.062

=== 2. Dependence of IN1 exceedance indicators (alpha = 0.1), mean over units ===
lag-1 (256 pulses) autocorrelation within bin: -0.007 (range -0.067 to 0.046)
correlation across bins at the same time: 0.045

=== 3. Leave-one-day-out range of headline effects (estimate on all 6 days; min..max over the 6 LOD0 fits) ===
NA-AR(4)/IN-AR(4) radius ratio (0.90): 0.891 LOD0 0.884..0.895
IN1 quintile Pfa spread (alpha 0.01): 1.75x LOD0 1.72..1.83x
single-pulse gain IN1 vs CA16 Pd=0.5: 6.96 dB LOD0 5.97.. 8.16
single-pulse gain IN1 vs CAlc Pd=0.5: 9.81 dB LOD0 8.94..11.54
single-pulse gain NA4 vs IN1 Pd=0.5: 1.07 dB LOD0 1.03.. 1.11
single-pulse gain NA4 vs IN4 Pd=0.5: 0.66 dB LOD0 0.58.. 0.75
single-pulse gain IN1 vs CA16 Pd=0.9: 7.62 dB LOD0 6.59.. 9.01
single-pulse gain IN1 vs CAlc Pd=0.9: 8.97 dB LOD0 8.10..10.62
single-pulse gain NA4 vs IN1 Pd=0.9: 2.19 dB LOD0 1.88.. 2.39
single-pulse gain NA4 vs IN4 Pd=0.9: 1.46 dB LOD0 1.43.. 1.52
eight-look dwell gain IN1 vs CAlc-sum Pd=0.5: 10.70 dB LOD0 9.86..12.38
eight-look dwell gain NA4 vs CAlc-sum Pd=0.5: 11.74 dB LOD0 11.01..13.00
per-day values of the single-pulse IN1 vs CAlc gain (Pd 0.5): 1993-11-07 3.5, 1993-11-08 8.4, 1993-11-09 13.8, 1993-11-10 13.3, 1993-11-11 12.2, 1993-11-18 11.4

=== 4. Fit diagnostics (all confirmatory units, training thirds) ===
|r| of IN-ARCP: median 0.960, max 0.980; clamp |r| <= 0.98 binding in 5 of 56 units
NA-AR(2) Nelder-Mead converged in 56 of 56 units; NA-AR(4) in 16 of 56 (median iterations 1500, cap 1500)
whole-unit time (all 13 procedures, GPU): median 71 s

=== 5. Second radar: Pfa/alpha by CNR class (mean over units with data), exact law, counts ===
class (dB) <-5 -5..0 0..5 5..10 10..20 >=20
test episodes 721095 265416 229845 68556 31870 338
IN1 a=0.1 1.03 1.03 1.00 0.83 0.38 0.00 units: 24 24 24 24 24 2
NA4 a=0.1 0.83 1.36 1.19 0.88 0.54 0.00 units: 24 24 24 24 24 2
CA16 a=0.1 1.07 1.05 0.96 0.58 0.15 0.00 units: 24 24 24 24 24 2
law a=0.1 1.04 1.00 0.95 0.86 0.68 0.34 (exact law, 3-decimal coverage from jku_summary)
IN1 a=0.01 1.07 1.06 0.95 0.52 0.08 0.00 units: 24 24 24 24 24 2
NA4 a=0.01 0.69 1.68 1.27 0.66 0.22 0.00 units: 24 24 24 24 24 2
CA16 a=0.01 1.14 1.05 0.78 0.19 0.02 0.00 units: 24 24 24 24 24 2
law a=0.01 1.10 1.00 0.90 0.70 0.50 0.20 (exact law, 3-decimal coverage from jku_summary)
```

Exploratory: sensitivity of NA-AR(4) to optimizer convergence (restart from the capped solution for up to 4,500 further Nelder-Mead iterations; `r10/explore_na4_convergence.py`):

```
units 56; converged at cap 1500: 16; converged after restart (+4500): 55
log-likelihood gain per training episode: median 3.13e-06, max 1.54e-04
1-alpha=0.9: radius ratio long/orig geo-mean 0.9989 (range 0.9790..1.0132); coverage orig 0.9019 long 0.9020
1-alpha=0.99: radius ratio long/orig geo-mean 0.9997 (range 0.9492..1.0327); coverage orig 0.9905 long 0.9906
```

14 Second radar: full results

```
units 24 (station x Rx x rotation); test episodes per CNR class {'<-5': 721095, '-5..0': 265416, '0..5': 229845, '5..10': 68556, '10..20': 31870, '>=20': 338}
mean CNR per class (dB): -12.9 -2.2 2.4 7.2 13.4 20.1
```

```
=== 1-alpha = 0.9 ===
J1 marginal coverage: IN1 0.9000 | IN4 0.9000 | NA1 0.8999 | NA4 0.9000 | G1 0.8974 | NA4G 0.8933 | U 0.8998 | RMS 0.9000 | CA16 0.9000 | CAlc 0.9001
J2 coverage by CNR class (mean over units):
IN1 0.897 0.897 0.900 0.917 0.962 1.000 | mean max class deviation 0.066
NA1 0.939 0.850 0.840 0.863 0.882 0.945 | mean max class deviation 0.067
NA4 0.917 0.864 0.881 0.912 0.946 1.000 | mean max class deviation 0.055
U 0.972 0.896 0.824 0.678 0.305 0.017 | mean max class deviation 0.629
CA16 0.893 0.895 0.904 0.942 0.985 1.000 | mean max class deviation 0.086
IN1 exact law 0.896 0.900 0.905 0.914 0.932 0.966 | MAE per unit x class 0.0104

=== 1-alpha = 0.99 ===
J1 marginal coverage: IN1 0.9900 | IN4 0.9900 | NA1 0.9900 | NA4 0.9901 | G1 0.9897 | NA4G 0.9846 | U 0.9899 | RMS 0.9900 | CA16 0.9900 | CAlc 0.9900
J2 coverage by CNR class (mean over units):
IN1 0.989 0.989 0.990 0.995 0.999 1.000 | mean max class deviation 0.009
NA1 0.996 0.983 0.979 0.984 0.989 1.000 | mean max class deviation 0.013
NA4 0.993 0.983 0.987 0.993 0.998 1.000 | mean max class deviation 0.009
U 1.000 0.997 0.991 0.959 0.734 0.245 | mean max class deviation 0.305
CA16 0.989 0.989 0.992 0.998 1.000 1.000 | mean max class deviation 0.010
IN1 exact law 0.989 0.990 0.991 0.993 0.995 0.998 | MAE per unit x class 0.0019

J3 detection (Swerling 1, Pfa 0.01): SCR gain vs CAlc at Pd=0.5 per CNR class, mean [95% CI]; closed form G_IN
G_IN(c) dB -0.79 -0.25 0.28 0.77 1.11 1.92
IN1 -1.45 [-1.79,-1.13] -1.00 [-1.19,-0.83] 0.02 [-0.23, 0.23] 1.57 [ 0.89, 2.20] 3.11 [ 1.97, 4.26] 5.78 [ 5.78, 5.78]
IN4 -1.42 [-1.57,-1.28] -0.85 [-0.99,-0.72] 0.52 [ 0.03, 0.96] 3.19 [ 2.27, 3.98] 6.78 [ 5.08, 8.34] 12.33 [12.33,12.33]
NA1 -2.10 [-2.65,-1.61] 0.65 [ 0.31, 0.93] 1.99 [ 1.06, 2.70] 4.21 [ 2.71, 5.51] 7.81 [ 5.26,10.08] 13.07 [13.07,13.07]
NA4 -1.26 [-1.38,-1.14] 0.35 [ 0.08, 0.64] 1.33 [ 0.78, 1.88] 3.61 [ 2.70, 4.44] 7.03 [ 5.50, 8.59] 12.78 [12.78,12.78]
U -5.48 [-6.02,-5.02] -2.19 [-2.65,-1.80] 0.22 [-0.21, 0.61] 4.11 [ 3.54, 4.65] 11.44 [10.93,11.95] 13.07 [13.07,13.07]
CA16 -0.62 [-0.64,-0.59] -0.49 [-0.55,-0.44] -0.47 [-0.50,-0.46] -0.54 [-0.57,-0.51] -0.61 [-0.67,-0.54] -0.85 [-0.85,-0.85]

J4 interference scenario C (Y-chirp hit rate 0.020; hit episodes 1920, clean 53520)
```

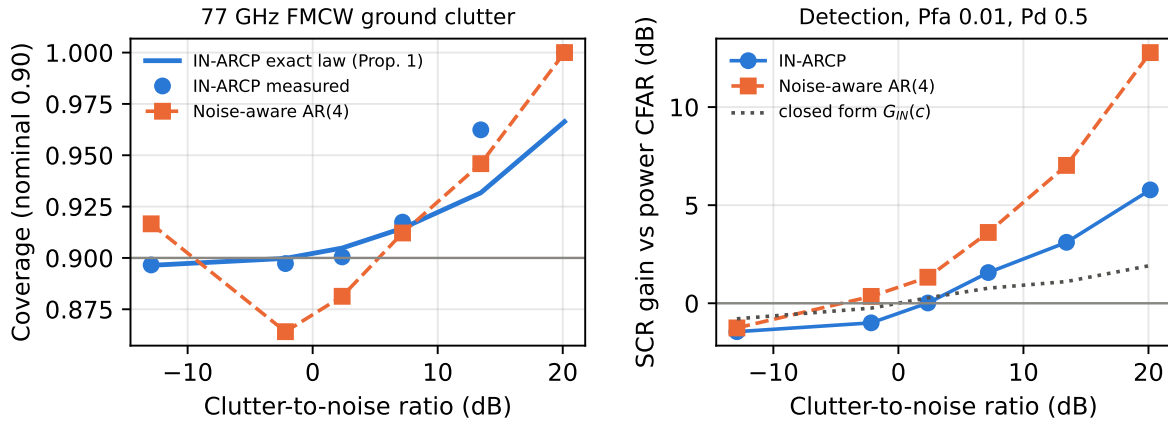


Figure 5: 77 GHz FMCW ground clutter: coverage by CNR class with the exact law, and SCR gain over the local power detector.

IN1	coverage 0.90: clean 0.901 hit 0.792 0.99: clean 0.990 hit 0.979
NA1	coverage 0.90: clean 0.828 hit 0.584 0.99: clean 0.964 hit 0.854
NA4	coverage 0.90: clean 0.870 hit 0.644 0.99: clean 0.980 hit 0.926
U	coverage 0.90: clean 0.761 hit 0.449 0.99: clean 0.950 hit 0.785
CA16	coverage 0.90: clean 0.900 hit 0.719 0.99: clean 0.989 hit 0.965